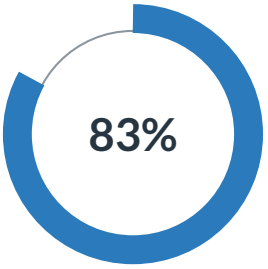
Results

Carter Feng



41.5
Out of 50 points

01:16:59
Time for this attempt

Your Answers:

1 0 / 0 points Fill in the Blank

At Seven Lakes High School, we believe that character matters and that academic integrity reflects who we are as learners and citizens. By signing this pledge, I commit to upholding the values of honesty, responsibility, and respect in all academic and personal endeavors.

My Academic Integrity Pledge:

- **I will complete all academic work honestly, without giving or receiving unauthorized assistance.**
- **I will not copy, plagiarize, or submit someone else's work as my own.**
- **I will not use unauthorized materials (notes, devices, online resources) during assessments.**
- **I will respect the integrity of others' work and will not aid others in academic dishonesty.**
- **I will not share or receive academic work, ideas, or answers electronically or in-person when individual work is expected or collaboration is not permitted.**
- **I will report any knowledge of cheating or violations of academic integrity to a trusted adult.**

I understand that academic dishonesty harms my character, undermines trust in our community, and may result in consequences outlined in the school's academic integrity policy.

Print your name to acknowledge the oath. Not printing your name will result in a zero for the assessment.

✓ Carter Feng

2 5 / 5 points Essay

Write the rightRightRotation method for a RedBlackTree class.

```
private RedBlackNode rightRightRotation(RedBlackNode g, RedBlackNode p
{
```

```
private RedBlackNode rightRightRotation(RedBlackNode g, RedBlackNode p)
{
    p.setColor(p.getColor() + 1);
    g.setColor(g.getColor() - 1);
    g.setRight(p.getLeft());
    p.setLeft(g);
    return p;
}
```

Correct

3 4.5 / 5 points Fill in the Blank

Create a red-black tree using the following values: 63, 43, 88, 72, 28, 78, 14, 45, 12, 85, 16, 76, 20 ,100, 35.
Delete the following values: 76, 72, 14, 12, 43.

What is the post order traversal of all the black nodes in the red-black tree?

Enter null when nothing is present.

Only enter value, not color.

Each space is one value, enter in the values in row major order.

✓ 16 ✗ 48 ✓ 28 ✓ 78 ✓ 100 ✓ 88

Correct Answer: **45**

✓ 63 ✓ null ✓ null ✓ null

4 5 / 5 points Matching

Create a red-black tree using the following values:63, 43, 88, 72, 28,

Delete the following values: 76, 72, 14, 12, 43.

What are the following nodes values when the 2nd rotation occurs?

What type of rotation is it?	✓ left - left
Red Parent Node	✓ 28
Red Child (or X) Node	✓ 14
Black Grandparent Node	✓ 43

5 2 / 5 points Essay

Write the simpleCaseDeletion method for a RedBlackTree class.

```
private RedBlackNode simpleCaseDeletion(RedBlackNode nodeToRemove, Red
{
```

```
private RedBlackNode simpleCaseDeletion(RedBlackNode nodeToRemove, RedBlackNoc
{
    nodeToRemove = child;
    child = null;
    return nodeToRemove;
}
```

Graded

6 0 / 5 points Essay

Write the leftLeftRestructure method for a RedBlackTree class.

```
private RedBlackNode leftLeftRestructure(RedBlackNode p, RedBlackNode
{
```

```
private RedBlackNode leftLeftRestructure(RedBlackNode p, RedBlackNode s, RedBl
{
    adjustmentLeft(p,s);
    recolor();
    leftLeftRotation(s,r);
    return p;
}
```

Incorrect

7 5 / 5 points Fill in the Blank

Create an AVL Tree.
Insert: 10 72 97 78 48 28 87 1 2 80 61 82 6 18 56

How many nodes of degree 1 are in the AVL tree?

✓ 0

8 5 / 5 points Fill in the Blank

Create an AVL Tree.
Insert: 10 72 97 78 48 28 87 1 2 80 61 82 6 18 56

What is the preorder traversal of all the nodes in the AVL tree?
Enter values in row major order.

✓ 72 ✓ 28 ✓ 2 ✓ 1 ✓ 10 ✓ 6
✓ 18 ✓ 56 ✓ 48 ✓ 61 ✓ 87 ✓ 80
✓ 78 ✓ 82 ✓ 97

9 5 / 5 points Fill in the Blank

Create an AVL Tree.
Insert: 10 72 97 78 48 28 87 1 2 80 61 82 6 18 56

What is the width of the AVL Tree?

✓ 6

10 5 / 5 points Matching

Create an AVL Tree.
Insert: 10 72 97 78 48 28 87 1 2 80 61 82 6 18 56

What are the following nodes values when the 4th rotation occurs?

What type of rotation is it? — ✓ left - right

Child Node	✓ 2
Parent Node	✓ 1
Ancestor Node	✓ 10

11 5 / 5 points Matching

Create an AVL Tree.
Insert: 10 72 97 78 48 28 87 1 2 80 61 82 6 18 56

What are the following nodes values when the 1st rotation occurs?

What type of rotation is it?	✓ right - right
Child Node	✓ 97
Ancestor Node	✓ 10
Parent Node	✓ 72